EXPERIMENTAL PAPER BLUEPRINT

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Programme & Institution:	MSc Data Science and Computational Intelligence | Softwarica College / Coventry University
Module & Assignment:	STW7072CEM / ST7072CEM Machine Learning Coursework CW (Regular)
Coursework Research Project:	**Machine Learning-Based Analysis of Daily Rainfall Patterns and Extreme Rainfall Events in the Kathmandu Valley**

Reference Paper Blueprint

- **Target Paper:** 'How accurately can satellite products (TMPA and IMERG) detect precipitation patterns, extremities, and drought across the Nepalese Himalaya?'
- **Citation:** Sharma et al. (2020)
- Paper link: https://doi.org/10.1029/2020EA001315

How to use this worksheet

Check the average length of comparable papers on a similar topic or in the same journal

1. Check the average length of comparable papers on a similar topic or in the same journal look for papers close to yours in methodology.

Comparable paper	Length (excl. references)
Paper 1: Lei et al. (2025), Water 17(12), 1776	~8,400 words
Paper 2: Sharma et al. (2020), Earth and Space Science 7(8), e2020EA001315	~7,900 words
Paper 3: Khanal et al. (2024), J. Hydrol. Meteorol. 12(1), 34-45	~4,200 words
Paper 4	—
Paper 5	—
Average	**~6,800 words**

2. Adjust the overall suggested length using that average.
3. Adjust the length of each section based on the papers you reviewed.
4. Check which sections are used in your field and edit the table accordingly:
    - Which sections from the table are present? Delete any that aren't.
    - Is there a separate literature review section?
    - Are results and discussion one section, or two?
    - Are discussion and conclusion one section, or two?
5. Compare the elements listed in each section against papers in your field:
    - Are all the listed elements present? Delete any that aren't.
    - Are they presented in the same order? If not, reorder the table.
6. Save the completed table below as its own document and this becomes your personal blueprint for future empirical papers.

Suggested overall length

- **Benchmark:** 6,000–10,000 words (without references)
- **Recommended range:** 7,000–9,000 words
- **Average length in my field:** ~7,900 words

Blueprint Table

Section	Element	Purpose	In my field?	My notes and ideas
Introduction*(800–1200 words)*	Importance of the topic	Introduce topic	✓ Yes	In the Nepalese Himalayas, **precipitation** controls glacier mass balance and triggers floods, landslides, and droughts. Gauges are sparse and situated in valley bottoms; validated satellite data are critical.
	Brief literature review	Field context	✓ Yes	Reviews TRMM TMPA 3B42V7 and **GPM IMERG** V06B; notes sparse prior evaluations in Nepal with few stations or short records.
	Research gap	Justification	✓ Yes	Lack of long-term (multi-decadal), nationwide comparison of TMPA and **IMERG** across complex Himalayan terrain covering daily scales, extremes, and multi-scalar drought.

Section	Element	Purpose	In my field?	My notes and ideas
	Research aim	Study aim	✓ Yes	Comprehensively assess the spatiotemporal **accuracy** of TMPA (3B42V7) and **IMERG** V06B against 142 rain gauges across Nepal over 2001–2018 for **precipitation** cycles, daily detection, extreme events (R10mm, R25mm, CDD, CWD), and drought (SPI).
	Main contributions	Novel insights	✓ Yes	18-year nationwide benchmark of **GPM** vs TRMM; proves **IMERG**'s daily superiority (CC=0.77) and enhanced drought monitoring (SPI-1 to SPI-12); identifies systematic orographic biases.
	Structure of paper	Paper organization	✓ Yes	Standard IMRAD: Sec 2 describes data/methods; Sec 3 presents annual, seasonal, daily, extreme, and drought results; Sec 4 discusses physical causes; Sec 5 concludes.
Literature Review*(optional)*	Key themes & Critical analysis	Literature review	✓ Yes	Integrated directly into Introduction and Discussion sections.
Theoretical / Methodological Framework*(200–500 words)*	Theories utilized	Framework	✓ Yes	Point-to-pixel **validation**; continuous metrics (CC, RMSE, ME, Bias, NSE); categorical contingency statistics (POD, FAR, ACC); ETCCDMI extreme **precipitation** indices; Standardized **Precipitation** Index (SPI).
Methodology*(1500–2000 words)*	Definition and justification	Methodology	✓ Yes	Point-to-pixel comparison avoids interpolation distortion in rugged relief; daily accumulation matching to 03:00 UTC gauge windows; quality screening (>=84% daily data/month).
	Background information	Study area	✓ Yes	Nepal (147,516 km²), elevation 60 m to 8,848 m across 5 physiographic regions; monsoon dominates (~80% in Jun–Sep); rain-shadow in trans-Himalayan valleys (Mustang/Manang).

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	Sample and sampling	Data sample	✓ Yes	142 rain gauge stations (141 **DHM** + 1 Pyramid AWS at 5,050 m asl) over 2001–2018 (18 years); TRMM 3B42V7 (0.25°) and **GPM IMERG** V06B Final Run (0.1°).
	Research tools and procedures	Tools & procedures	✓ Yes	MATLAB, ArcGIS, NASA Giovanni. Extracted satellite data to station coordinates, computed continuous/categorical metrics, derived extreme indices and SPI-1, 4, 6, 12.
	Data analysis techniques	Analysis techniques	✓ Yes	Continuous statistical metrics across timescales; contingency metrics (POD, FAR, ACC); frequency of R10mm, R25mm, CDD, CWD; cumulative drought frequency and Pearson correlation for SPI.
Results*(2000–3000 words)*	Presentation of main results	Results	✓ Yes	(1) Both capture annual cycle but underestimate monsoon; (2) **IMERG** superior on daily scale (CC=0.77 vs 0.70; RMSE=4.30 vs 4.79 mm/day; POD>80%); (3) TMPA slightly more consistent on monthly scale (ME=-8.54mm); (4) Extremes: both overestimate R10mm/CWD and underestimate R25mm/CDD; (5) Drought: **IMERG** superior in capturing drought frequency (CC=0.95, RMSE=0.32).
	Tables and figures	Evidence	✓ Yes	Station distribution map; monthly annual cycle; **spatial** maps of mean annual **precipitation** and seasonal bias; daily time series & anomalies; **spatial** POD/FAR/ACC maps; extreme event maps; **temporal** SPI plots & cumulative drought frequency; scatter plots.
Discussion*(1200–1800 words)*	Restatement of result	Restatement	✓ Yes	Reasserts **IMERG**'s superiority at daily and drought timescales, and TMPA's marginal edge on monthly

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				accumulations; highlights satellite limitations over steep topography.
	Comparison with literature	Literature context	✓ Yes	Matches findings in Tibetan Plateau and China showing **IMERG** improves daily detection and reduces false alarms due to DPR; confirms difficulty of satellite radar in capturing shallow orographic clouds over high terrain.
	Explanation of findings	Explanations	✓ Yes	Higher **spatial** resolution (0.1°) and Dual-frequency **Precipitation** Radar enhance daily detection and solid/light **precipitation** capture in winter. Underestimation at orographic hotspots (e.g. Lumle >3000mm) occurs because microwave retrieval associates **rainfall** rate with echo-top height, misidentifying shallow warm orographic clouds.
	Interpretation of findings	Interpretation	✓ Yes	**IMERG** is a dependable alternative for daily hydrometeorological modeling and drought monitoring in data-scarce Himalayan basins, but requires bias correction before use in flash flood forecasting.
Conclusion*(600–800 words)*	Restatement of aim	Main aim	✓ Yes	Evaluated spatiotemporal **accuracy** of TMPA and **IMERG** against 142 gauge stations across Nepal (2001–2018) for **precipitation** patterns, extremes, and drought.
	Recap of key findings	Key findings	✓ Yes	**IMERG** outperforms TMPA at daily timescale (CC=0.77) and drought monitoring (SPI CC=0.95); TMPA is slightly better for monthly sums; both products underestimate heavy orographic **precipitation**.

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	Practical implications	Stakeholder impact	✓ Yes	**IMERG** recommended for daily hydrological modeling, disaster risk reduction, and drought monitoring; TMPA suitable for monthly water balance assessments.
	Limitations	Limitations	✓ Yes	Gauge scarcity at high altitudes (>3000m); valley-bottom gauge bias; potential GPCC gauge adjustment dependency.
	Suggestions for future research	Future research	✓ Yes	Investigate sub-daily (hourly/half-hourly) extreme **rainfall**; develop elevation-dependent bias correction algorithms.